

2. Architecture Diagram

3. Login & Authentication Flow

	%	A u t h C o n t e x t c h e c k s :	%			
	%	1 . b a s e 4 4 . a u t h . m e ()	%			
	%	2 . H M A C t o k e n (l o c a l)	%			
	%	3 . L o c a l t e s t m o d e	%			
	% % % % % % % % % % % % , % % % % % % % % % %					
		%				
	% %					
	%	A u t h e n t i c a t e d ?	%			
	% % % % % % % , % % % % % % % % , % % % % % % % %					
		Y E S %	% N O			
% % % % % % % % % % % % % % % % % %		% % % % % % % % % % % % % % % % % %	%			
%			%			
%¼			%¼			
% % % % % % % % % % % % % % % %		% % % % % % % % % % % % % % % % % %	%			
%	R o l e C h e c k	%	L o g i n P a g e	%		
%	c r e a t o r ?	%	- E x c h a n g e	%		
%	a d m i n ?	%	- C u s t o m e r	%		
%	s a r r a a f ?	%	- G o o g l e O A u t h	%		
%	e m p l o y e e ?	%	% % % % % % % % % % % % % % % % % %			
%	c u s t o m e r ?	%	%			
% % % % , % % % % % , % % % % % %		%				
%	%	% % % % % % % % % % % % % % % % % %	%¼ % % % % % %			
%	%	%	B a c k e n d : l o g i n	%		
%	%	%	- V e r i f y p a s s w o r d	%		
%	%	%	- R a t e l i m i t c h e c k	%		
%	%	%	- I s s u e H M A C t o k e n	%		
%	%	%	- R e s o l v e e x c h a n g e	%		
%	%	% % % % % % % % % % % % % % % % % , % % % % % % % %				
%	%	%	%			
%¼	%¼		%¼			
% % % % % % % % % % % % % % % %	% % % % % % % % % % % % % % % %	% % % % % % % % % % % % % % % % % %				
%	B u i l d e r	%	S a r r a a f	%	C u s t o m e r P a n e l	%
%	P a n e l	%	P a n e l	%	/ c u s t o m e r / *	%
%	/ b u i l d e r	%	/ d a s h b o a r d	% % % % % % % % % % % % % % % % % %		
% % % % % % % % % % % %	% % % % % % % % % % % %					

4. Data Flow Diagram

% % % % % % % % % %	% % % % % % % % % %	% % % % % % % % % %
% User % % % % % % % %	% React % % % % % % % %	% React % % % % % % % %
% Action % % % % % % % %	% Component % % % % % % % %	% Query % % % % % % % %
% % % % % % % % % %	% (use State) % % % % % % % %	% (use Sarraaf) % % % % % % % %
% % % % % % % % % %	% % % % % % % % % %	% % % % % , % % % % % % % %
		%
	% % % % % % % % % %	% % % % % % % % % %
	% supabaseApi.js %	%
	% callSupabase(action , data)%	%
	% - Field mapping %	%
	% - Auth token injection %	%
	% % % % % % % % % %	% % % % % % % % % %
	%	%
	% % % % % % % % % %	% % % % % % % % % %
	% Backend Function (Deno) %	%
	% base44.functions.invoke %	%
	% - Session verification %	%
	% - Tenant scoping %	%
	% - Amount validation %	%
	% % % % % % % % % %	% , % % % % % % % % % %
	%	%
	% % % % % % % % % %	% % % % % % % % % %
	% Supabase (service_role) %	%
	% - Insert / Update / Select %	%
	% - RPC (atomic_ops) %	%
	% - Views (balances) %	%
% % % % % % % % % %	% % % % % % % % % %	% % % % % % % % % %

5. Database Schema & Relationships

Tables (54+):

users
exchanges
customers
documents
transfers
conversions
treasury_transactions
bank_accounts
bank_transactions
journal_entries
ledger_entries
rates
idempotency_keys
employees
origins
safes
rules
feature_settings
app_settings
audit_logs
wallet_transactions
charge_orders
online_requests
online_products
wallets
wallet_withdrawal_requests
wallet_topup_requests
wallet_bans
customer_notifications
customer_transfer_requests
customer_links
email_otps
user_notifications
user_permissions
user_feedback
support_tickets
support_messages
error_reports
bank_gateways
payment_transactions
chat_groups
chat_group_members
chat_messages
message_reports
sellers
packages
pricing_plans
plan_quotas
quota_change_logs
referrals
agency_links
backup_records
rate_settings
text_labels

Key Relationships

```
users (id) % % % % % exchanges (owner_user_id)
% % % customers (user_id)
% % % wallets (user_id)

exchanges (exchange_code) % % % % % documents
```

```

                                % % %   t r a n s f e r s
                                % % %   c u s t o m e r s
                                % % %   r a t e s
                                % % %   b a n k _ a c c o u n t s
                                % % %   s a f e s
                                % % %   a l l   t e n a n t - s c o p e d   t a b l e s

d o c u m e n t s   ( i d ) % % % %   l e d g e r _ e n t r i e s   ( d o c u m e n t _ i d )
d o c u m e n t s % % % %   t r a n s f e r s   ( t r a n s f e r _ i d )

w a l l e t s   ( i d ) % % % %   w a l l e t _ t r a n s a c t i o n s   ( w a l l e t _ i d )
w a l l e t s % % % %   c h a r g e _ o r d e r s   ( w a l l e t _ i d )
w a l l e t s % % % %   w a l l e t _ w i t h d r a w a l _ r e q u e s t s   ( w a l l e t _ i d )

c h a t _ g r o u p s   ( i d ) % % % %   c h a t _ g r o u p _ m e m b e r s
c h a t _ g r o u p s % % % %   c h a t _ m e s s a g e s
c h a t _ m e s s a g e s % % % %   m e s s a g e _ r e p o r t s

```

6. RLS & Security Diagram

```

% % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %
%                               S U P A B A S E   R L S   L A Y E R                               %
%                                                                                               %
%   % % % % % % % % % % % % % %   % % % % % % % % % % % % % %   % % % % % % % % % % % % % %   %
%   %   a n o n   k e y   %   %   a u t h e d   k e y   %   %   s e r v i c e   % % % % % % % % % %
%   %   ( f r o n t e n d ) %   %   ( a u t h e d )   %   %   r o l e   % % % % % % % % % %
%   % % % % % % % % % % % % % % % %   % % % % % % % % % % % % % %   % % % % % % % % % % % % % %
%   %                               %                               %                               %
%   %                               %                               %                               %
%   % % % % % % % % % % % % % %   % % % % % % % % % % % % % %   % % % % % % % % % % % % % %
%   %   R E A D   O N L Y %   %   R E A D   O N L Y %   %   F U L L   A C C E S S % % % % % % % % % %
%   %   5   p u b l i c   %   %   5   p u b l i c   %   %   ( b y p a s s e s   % % % % % % % % % %
%   %   t a b l e s   %   %   t a b l e s   %   %   R L S )   % % % % % % % % % %
%   % % % % % % % % % % % % % %   % % % % % % % % % % % % % %   % % % % % % % % % % % % % %
%
%   P u b l i c   ( r e a d - o n l y ) :   r u l e s ,   f e a t u r e _ s e t t i n g s ,
%   p r i c i n g _ p l a n s ,   o n l i n e _ p r o d u c t s
%
%   A l l   o t h e r   t a b l e s :   D E N Y   A L L   f o r   a n o n / a u t h e d
%   ( n o   p o l i c y   =   d e n y ;   s e r v i c e _ r o l e   b y p a s s e s )
% % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %

```

- Application Security:
- HMAC-SHA256 tokens (12h TTL, signed with SERVICE_KEY)
 - PBKDF2 password hashing (100k iterations)
 - Rate limiting (5 fails / 15 min = lockout)
 - Tenant isolation (exchange_code scoping)
 - Audit logging (all CRUD + auth events)

7. Financial Logic Diagram

```

% % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %
%                               D O U B L E - E N T R Y   L E D G E R                               %
%                                                                                               %
%   E v e r y   t r a n s a c t i o n :                                                                                               %
%   % % % % % % % % % % % % % %   % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %
%   %   D o c u m e n t % % % % % % % % % %   L e d g e r   L i n e s % % % % % % % % % % % % % % % % % % % % % % % %
%   %   ( h e a d e r ) %   %   ( d e b i t + c r e d i t ) % % % % % % % % % % % % % % % % % % % % % % % %
%   % % % % % % % % % % % % % %   % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %
%
%   B a l a n c e   c h e c k :   s u m ( d e b i t )   =   s u m ( c r e d i t )   p e r   c u r r e n t   %
%                                                                                               %
%   F l o w :                                                                                               %
%   1 .   U s e r   s u b m i t s   ( d a d o g e r e f t / t r a n s f e r / e t c ) %
%   2 .   p o s t D o c u m e n t   a c t i o n %

```

```

%      3 .   V a l i d a t e   d o u b l e - e n t r y   b a l a n c e                               %
%      4 .   I d e m p o t e n c y   c h e c k   ( c l i e n t _ o p e r a t i o n _ i d )           %
%      5 .   G e n e r a t e   d o c _ n u m b e r   ( D O C - 0 0 0 0 0 1 )                     %
%      6 .   I n s e r t   d o c u m e n t   h e a d e r                                       %
%      7 .   I n s e r t   l e d g e r _ e n t r i e s   ( d e b i t   +   c r e d i t )           %
%      8 .   I f   l i n e s   f a i l   ! '   r o l l b a c k   d o c u m e n t                   %
%      9 .   R e c o r d   i d e m p o t e n c y   k e y                                       %
%     1 0 .   A u d i t   l o g                                                                %
%                                                                                               %
%   W a l l e t   O p e r a t i o n s :                                                         %
%   -   R e a d   w a l l e t . b a l a n c e                                                   %
%   -   C h e c k   n e w B a l a n c e   > =   0                                             %
%   -   U P D A T E   . . .   W H E R E   b a l a n c e   =   c u r r e n t B a l a n c e         %
%       ( o p t i m i s t i c   l o c k   p r e v e n t s   c o n c u r r e n t   o v e r s p e n d ) %
% % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % % %

```

8. File Dependency Map

```

src/main.jsx
  % % %   s r c / A p p . j s x   ( R o u t e r   +   P r o v i d e r s )
  % % %   s r c / l i b / A u t h C o n t e x t . j s x
  % % %   s r c / l i b / i 1 8 n . j s x   ( L a n g u a g e   P r o v i d e r )
  % % %   s r c / l i b / U p g r a d e C o n t e x t . j s x
  % % %   s r c / c o m p o n e n t s / A p p G a t e . j s x
  % % %   s r c / c o m p o n e n t s / P r o t e c t e d R o u t e . j s x
  % % %   s r c / c o m p o n e n t s / A d m i n R o u t e . j s x
  % % %   s r c / c o m p o n e n t s / s a r r a a f / L a y o u t . j s x
  % % %   s r c / c o m p o n e n t s / C u s t o m e r L a y o u t . j s x
  % % %   s r c / p a g e s / *   ( 5 0 +   p a g e s )
  % % %   s r c / l i b / s u p a b a s e A p i . j s
  % % %   s r c / l i b / s u p a b a s e C l i e n t . j s
  % % %   s r c / l i b / o f f l i n e Q u e u e . j s
  % % %   s r c / l i b / u s e S a r r a a f . j s

```

```

src/lib/useSarraaf.js (React Query hooks)
  % % %   s r c / l i b / s u p a b a s e A p i . j s   ( c a l l S u p a b a s e )

```

```

base44/functions/supabase/entry.ts (backend)
base44/shared/supabase_schema.sql (DDL)
base44/shared/rls_policies.sql (RLS)
base44/entities/*.jsonc (schemas)

```

Key shared libraries:

```

src/lib/rbac.js (role-based access control)
src/lib/ledgerApi.js (double-entry helpers)
src/lib/builderAccess.js (builder panel access)
src/lib/creatorAuth.js (creator role)
src/lib/features.js (feature catalog)
src/lib/permissions.js (permission system)

```

9. Base44 Dependencies

Files and components that depend on Base44 platform:

```

@base44/sdk | 0.8.51 | Base44 client SDK — auth, entities, integrations
@base44/vite-plugin | 1.0.42 | Vite plugin for Base44 platform
src/api/base44Client.js | — | Pre-initialized base44 client instance
src/components/ProtectedRoute.jsx | — | Auth guard using base44.auth.isAuthenticated()
src/components/AdminRoute.jsx | — | Admin route guard using base44 auth
src/components/AppGate.jsx | — | App-level gate with base44 auth context
src/lib/AuthContext.jsx | — | Auth context with base44.auth fallback
src/components/ui/* | — | shadcn/ui components (via @base44/vite-plugin)
index.html | — | Base44 runtime script injection in <head>
vite.config.js | — | Vite config with @base44/vite-plugin
base44/functions/supabase/entry.ts | — | Backend function (Deno runtime)
base44/shared/supabase_schema.sql | — | Database schema SQL

```

base44/shared/rls_policies.sql | — | RLS policies SQL
base44/config.jsonc | — | Base44 app configuration
base44/entities/*.jsonc | — | Entity schema definitions
src/lib/seedBase44Entities.js | — | Entity seeding utility

Migration notes:

- `base44.auth.*!` Replace with Supabase Auth or custom JWT
- `base44.entities.*!` Replace with `supabaseApi.js` (already done)
- `base44.functions.invoke!` Replace with direct Supabase calls or Edge Functions
- `base44.integrations.Core.*!` Replace with direct API calls
- `@base44/vite-plugin!` Remove from `vite.config.js`
- `index.html <head>!` Remove Base44 runtime script injection

10. Supabase Dependencies

Real Supabase connections, tables, and services:

@supabase/supabase-js | 2.109.0 | Supabase JavaScript client
src/lib/supabaseClient.js | — | Direct Supabase client (VITE_ env vars)
src/lib/supabaseApi.js | — | Supabase API layer (entity CRUD + field mapping)
src/lib/supabaseDB.js | — | Direct Supabase interaction module
src/lib/supabaseAuth.js | — | Supabase auth helpers
base44/functions/supabase/entry.ts | — | Backend function using `service_role` key
VITE_SUPABASE_URL | — | Frontend Supabase URL (public, safe)
VITE_SUPABASE_KEY | — | Frontend Supabase anon key (public, RLS-protected)
SUPABASE_URL | — | Backend Supabase URL (server-side)
SUPABASE_KEY | — | Backend Supabase anon key (server-side)
SUPABASE_SERVICE_KEY | — | Backend service role key (bypasses RLS)
Tables (54+) | — | See Supabase Structure section
RLS Policies | — | See RLS section — defense in depth
Auth (OTP) | — | Supabase built-in email OTP for verification
Storage | — | Public bucket for file uploads (receipts)

Tables (54+): `users`, `exchanges`, `customers`, `documents`, `transfers`, `conversions`, `treasury_transactions`, `bank_accounts`, `bank_transactions`, `journal_entries`, `ledger_entries`, `rates`, `idempotency_keys`, `employees`, `origins`, `safes`, `rules`, `feature_settings`, `app_settings`, `audit_logs`, `wallet_transactions`, `charge_orders`, `online_requests`, `online_products`, `wallets`, `wallet_withdrawal_requests`, `wallet_topup_requests`, `wallet_bans`, `customer_notifications`, `customer_transfer_requests`, `customer_links`, `email_otps`, `user_notifications`, `user_permissions`, `user_feedback`, `support_tickets`, `support_messages`, `error_reports`, `bank_gateways`, `payment_transactions`, `chat_groups`, `chat_group_members`, `chat_messages`, `message_reports`, `sellers`, `packages`, `pricing_plans`, `plan_quotas`, `quota_change_logs`, `referrals`, `agency_links`, `backup_records`, `rate_settings`, `text_labels`

Views: `v_treasury_balances`, `v_bank_balances`, `v_customer_balances`, `v_wallet_balances`, `v_agent_balances`, `v_profit_loss`, `v_currency_balances`, `v_exchange_summary`, `v_transactions`

RPC Functions: `post_transfer_transaction`, `reverse_document_transaction`, `edit_document_transaction`, `get_account_balance`

11. Migration from Base44

- Step 1: Replace `base44.auth` with Supabase Auth or custom JWT
- Step 2: Replace `base44.entities` with `supabaseApi.js` (already done)
- Step 3: Replace `base44.functions.invoke` with direct Supabase calls
- Step 4: Replace `base44.integrations` with direct API implementations
- Step 5: Remove `@base44/vite-plugin` from `vite.config.js`
- Step 6: Remove Base44 runtime script from `index.html <head>`
- Step 7: Deploy backend function as Supabase Edge Function
- Step 8: Set environment variables (`SUPABASE_URL`, `SUPABASE_SERVICE_KEY`)
- Step 9: Test all CRUD operations via `supabaseApi.js`

Step 10: Verify RLS policies are active

12. Rebuild from Scratch

Step 1: Create new Vite + React project

Step 2: Install dependencies (see package.json)

Step 3: Set up Tailwind CSS + shadcn/ui

Step 4: Create Supabase project + run schema SQL

Step 5: Apply RLS policies SQL

Step 6: Create backend function (copy entry.ts)

Step 7: Set secrets (SUPABASE_URL, SUPABASE_SERVICE_KEY, CREATOR_EMAIL, CREATOR_PASSWORD)

Step 8: Copy source files (src/*)

Step 9: Update routing (App.jsx)

Step 10: Test auth + CRUD + financial operations

13. Frontend -> API -> Backend -> Database

[illegible]

Token flow:

```
localStorage (auth_token) !' callSupabase() !' entry.ts verifyToken
```

14. Routes (React Router)

```

/ | Welcome | Public
/setup-database | SetupDatabase | Public
/login | Login | Public
/register | Register | Public
/forgot-password | ForgotPassword | Public
/reset-password | ResetPassword | Public
/dashboard | Dashboard | Sarraaf
/customers | Customers | Sarraaf
/customers/:id | CustomerDetail | Sarraaf
/dadogereft | Dadogereft | Sarraaf
/trade | Trade | Sarraaf
/transfers/:direction | Transfer | Sarraaf
/cash | Cash | Sarraaf
/bank | Bank | Sarraaf
/currencies | Currencies | Sarraaf
/reports | Reports | Sarraaf
/profit-loss | ProfitLoss | Sarraaf
/settings | Settings | Sarraaf
/transfer-book | TransferBook | Sarraaf
/journal | Journal | Sarraaf
/general-ledger | GeneralLedger | Sarraaf
/journal-entry | JournalEntry | Sarraaf
/balance-sheet | BalanceSheet | Sarraaf
/reconciliation | Reconciliation | Sarraaf
/convert | CurrencyConvert | Sarraaf
/online-products | OnlineProducts | Sarraaf

```

/payment-gateway | PaymentGateway | Sarraaf
/agents | Agents | Sarraaf
/debtors | Debtors | Sarraaf
/notifications | Notifications | Sarraaf
/agency-network | AgencyNetwork | Sarraaf
/chat | ChatGroups | Sarraaf
/chat/group/:id | ChatGroupDetail | Sarraaf
/sarraaf-market | SarraafMarket | Sarraaf
/sarraaf-chat | SarraafChat | Sarraaf
/backup | Backup | Sarraaf
/support | SarraafSupport | Sarraaf
/builder | BuilderPanel | Creator/Admin
/main-settings | MainSettings | Creator/Admin
/customer | CustomerHome | Customer
/customer/wallet | CustomerWallet | Customer
/customer/recharge | CustomerRecharge | Customer
/customer/games | CustomerGames | Customer
/customer/transfer | CustomerTransfer | Customer
/customer/withdraw | CustomerWithdraw | Customer
/customer/support | CustomerSupport | Customer
/customer/account | CustomerAccount | Customer
/customer/orders | CustomerOrders | Customer
/customer/messages | CustomerMessages | Customer
/customer/notifications | CustomerNotifications | Customer
/customer/profile | CustomerProfile | Customer
/customer/chat | CustomerChat | Customer
/receipt | Receipt | Authenticated

15. Entities

Entity schemas defined in base44/entities/*.jsonc:

User
SarraafProfile
Customer
Wallet
Transaction
Transfer
Agent
Currency
BankAccount
BankGateway
Safe
OnlineProduct
OnlineOrder
ChatGroup
GroupMember
GroupMessage
MessageReport
WalletTransaction
WalletTopupRequest
WalletWithdrawalRequest
WalletBan
CustomerTransferRequest
CustomerNotification
CustomerLink
PaymentTransaction
PaymentRecord
AuditLog
AppSetting
FeatureSetting
PricingPlan
PlanQuota
QuotaChangeLog
UserPermission
UserNotification
UserFeedback

ErrorReport
SupportTicket
SupportMessage
Announcement
Referral
AgencyLink
Seller
Package
RateSetting
TextLabel
BackupRecord
BuilderAccess
CustomerChat
LedgerEntry

16. Security Model

- HMAC-SHA256 stateless session tokens (12h TTL)
- PBKDF2 password hashing (100k iterations, SHA-256)
- Rate limiting: 5 failed logins in 15 min = lockout
- Tenant isolation: exchange_code scoping on all exchange-scoped tables
- RLS: defense in depth — anon key can only read public tables
- Service role: bypasses RLS (backend only, never exposed)
- Google OAuth: creates customer with placeholder mobile
- Audit logging: all login, CRUD, password change events
- No secrets in frontend: VITE_ vars are public, RLS-protected
- Idempotency keys: prevent duplicate financial operations

17. Financial Logic

- Double-entry ledger: every transaction posts debit + credit lines
- Atomic document posting: documents + ledger_entries in single operation
- Idempotency: client_operation_id prevents duplicate operations
- Wallet operations: optimistic-lock balance updates (eq on current balance)
- Transfer posting: RPC post_transfer_transaction (atomic)
- Document reversal: RPC reverse_document_transaction
- Document edit: RPC edit_document_transaction (soft-delete old + insert new)
- Balances: SQL views (v_treasury_balances, v_bank_balances, v_customer_balances, etc.)
- Withdrawal: wallet deduction + withdrawal request + wallet_transaction
- Customer order: product price validation + wallet deduction + order creation

18. PWA & Offline

- Service Worker: public/sw.js (needs update for full PWA)
- Manifest: public/manifest.json
- Offline Queue: src/lib/offlineQueue.js (queues writes, syncs on reconnect)
- Cache: localStorage-based caching for offline reads
- Sync Status: src/components/SyncIndicator.jsx
- Keep Alive: src/lib/useKeepAlive.js (prevents Supabase inactivity pause)

19. Environment Variables (no secrets)

Frontend (public, safe — RLS protected):

VITE_SUPABASE_URL — Supabase project URL

VITE_SUPABASE_KEY — Supabase anon key (public, RLS-protected)

Backend (server-side, never exposed):

SUPABASE_URL — Supabase project URL
SUPABASE_KEY — Supabase anon key
SUPABASE_SERVICE_KEY — Service role key (bypasses RLS)
CREATOR_EMAIL — Creator account email
CREATOR_PASSWORD — Creator account password
BASE44_APP_ID — Base44 app ID (for SDK fallback)

NOTE: All secrets are set via Base44 Secrets dashboard, not in code.

20. Build & Deployment

Build: vite build (Vite + React)
Deploy: Base44 platform (auto-deploy on push)
Backend: Deno runtime (base44/functions/supabase/entry.ts)
Database: Supabase PostgreSQL (managed)
Storage: Supabase Storage (public bucket)
Mobile: Base44 native build (iOS/Android)
Custom Domain: Configure in Base44 dashboard

This PDF was generated as an emergency backup snapshot.

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Project: پښتو پښتو» پښتو پښتو (Hesab Sarraf)

Version: v01 — Emergency Backup

No secrets, passwords, or API keys are included in this document.

Use the ZIP backup for source code and the Supabase backup for database structure.